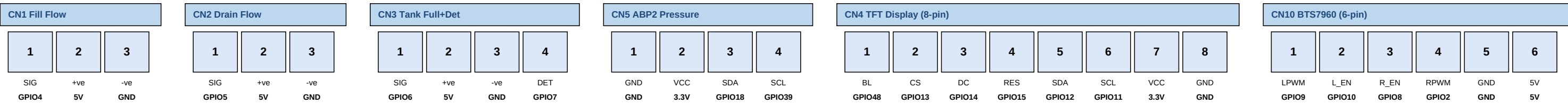
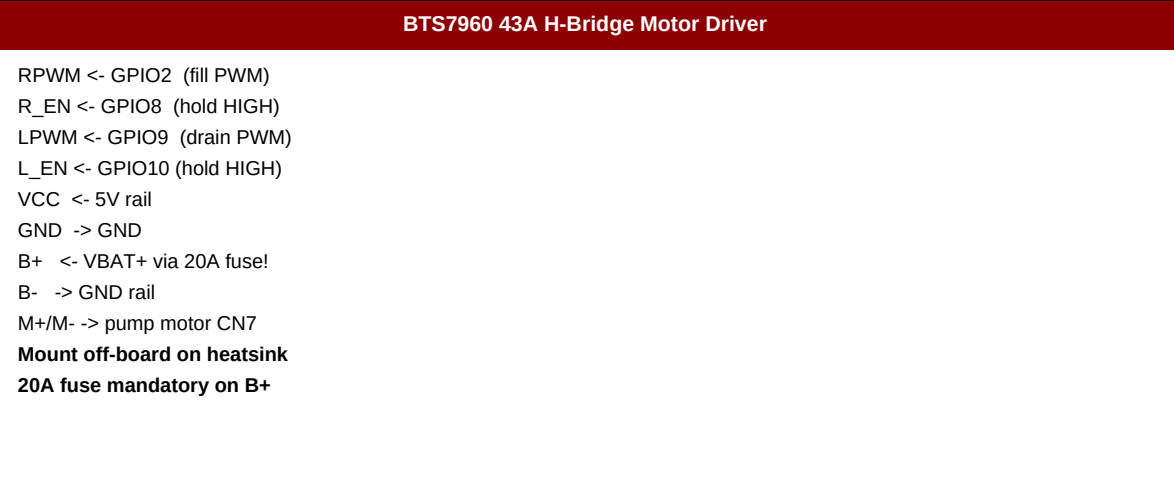
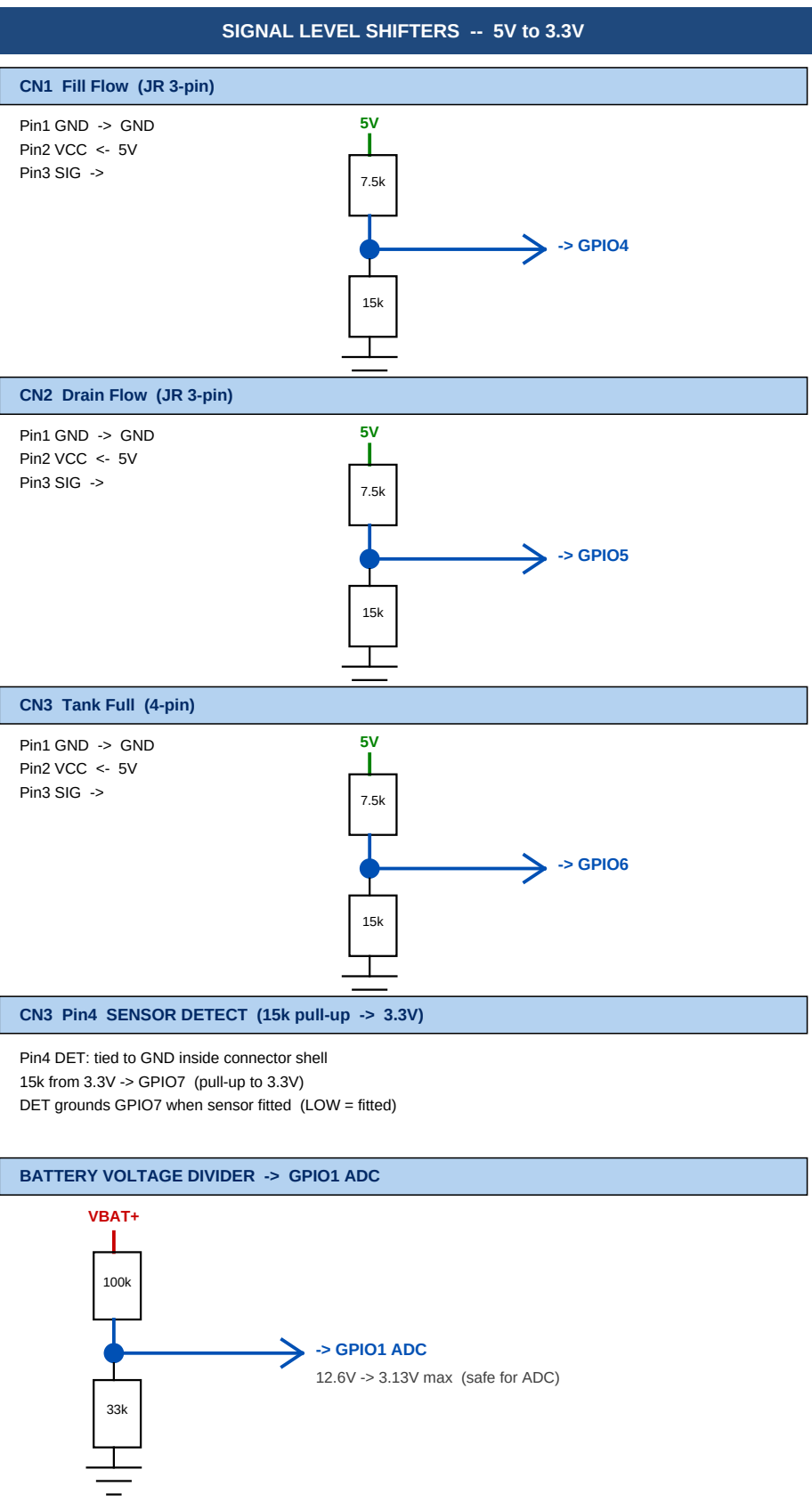
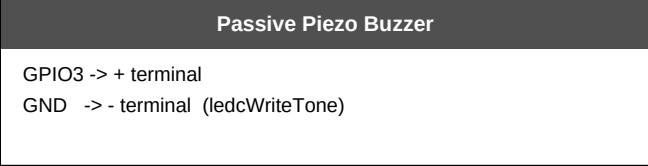


MCP Fuel Station v2 - Circuit Diagram

ESP32-S3 N16R8 | 3S LiPo | MP1584EN 5V | Pololu SV Mini | BTS7960 | ST7796S 3.5" IPS | DS3231 RTC | Honeywell ABP2 | GT911 Touch | KY-040 Encoder



1. Pre-set MP1584EN to 5.00V with multimeter BEFORE connecting to board.
2. Remove R3 from encoder PCB - eliminates SW pull-up holding Pololu A HIGH.
3. Remove DS3231 charge resistor (next to 1N4148 on back) before fitting CR2032.
4. BTS7960 must be mounted off-board on aluminium heatsink - gets hot under load.
5. 20A inline fuse is mandatory on VBAT+ to BTS7960 B+ terminal.
6. Test all power rails with multimeter before inserting ESP32.
7. Encoder: power from 3.3V ONLY - NOT 5V. Avoids 5V signals on ESP32 GPIO.
8. I2C0 (Wire0): DS3231=0x68, ABP2=0x28. I2C1 (Wire1): GT911=0x5D. No address conflicts.
9. GT911: Wire1 SDA=GPIO47, SCL=GPIO41, INT=GPIO42. No RST wired - INT floats HIGH at power-up -> addr 0x5D.
10. Poll GT911 unconditionally every 50ms - do NOT gate on INT (stays LOW after init without RST).
11. Display: ST7796S 3.5" IPS 480x320. TFT_eSPI: ST7796_DRIVER, USE_FSPI_PORT=1, setRotation(1).

GPIO41/42/47 now assigned